THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The smallest number by which 98 must be multiplied to get a perfect square is: 1

(a) 4 **(b)** 3 **(c)** 2 **(d)** 1

2. The value of $2^9 - 9^2$ is: 1

(a) 0 **(b)** -431 **(c)** 593 **(d)** 431

3. In 2403_5, the value (in decimal) contributed by the digit 4 in position 2 from the left is: 1

(a) 500 **(b)** 20 **(c)** 100 **(d)** 29

4. A quadrilateral whose diagonals bisect each other and are equal is a: 1

(a) rhombus **(b)** kite **(c)** rectangle **(d)** trapezium

5. The value of 83 - 38 is: 1

(a) 36 **(b)** 54 **(c)** 45 **(d)** 135

6. The smallest number by which 250 must be divided to get a perfect cube is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 4

7. $2^{-1} + 3^{-1}$ equals: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{1}{6}$

8. The decimal value of 11011_2 is: 1

(a) 33 **(b)** 27 **(c)** 25 **(d)** 29

9. The angles of a quadrilateral are x°, $(x+20)^\circ$, $(x+40)^\circ$ and $(x+60)^\circ$. The value of x is: 1

(a) 80 **(b)** 60 **(c)** 70 **(d)** 40

10. The digit that replaces * in 6*81 so that the number is divisible by 11 is: 1

(a) 5 **(b)** 3 **(c)** 2 **(d)** 8

11. The ones digit of 238^2 is: 1

(a) 2 **(b)** 4 **(c)** 6 **(d)** 8

12. $\dfrac{5^7 \times 5^3}{5^6}$ equals: 1

(a) 5^{10} **(b)** 5^{15} **(c)** 5^4 **(d)** 5^{16}

13. The binary form of 54 is: 1

(a) 110101 **(b)** 110110 **(c)** 011011 **(d)** 110111

14. The adjacent sides of a parallelogram are in the ratio 2:3 and its perimeter is 60 cm. The longer side is: 1

(a) 18 cm **(b)** 12 cm **(c)** 30 cm **(d)** 36 cm

15. The smallest digit that replaces * in 71*4 so that the number is divisible by 6 is: 1

(a) 2 **(b)** 0 **(c)** 1 **(d)** 5

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Evaluate $\sqrt{\frac{81}{256}}$. 2

17. Write $(6 \times 10^8) \div (2 \times 10^3)$ in scientific notation. 2

18. Subtract $11010_2 - 1001_2$ and give the answer in binary. 2

19. The exterior angles at three vertices of a quadrilateral are $70°$, $95°$ and $105°$. Find the fourth exterior angle and the fourth interior angle. 2

20. Find 842 - 248 for the 3-digit number 842 and check whether it is divisible by 99. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Express 54 as the sum of three consecutive natural numbers. 3

22. Find the smallest number by which 2646 must be multiplied to make it a perfect square. Find the square root of the product. 3

23. Simplify $\frac{3^7 \times 2^6}{3^5 \times 2^3}$ and find its value. 3

24. Convert 1432_5 to base 8 by first converting to decimal. 3

OR

The angles of a quadrilateral are $x°$, $(x+8)°$, $(x+16)°$ and $(x+24)°$. Find all the angles. 3

25. Find the smallest 4-digit number divisible by both 14 and 21. 3

OR

Find the smallest number by which 10976 must be divided to make it a perfect cube. Find the cube root of the quotient. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. The diagonals of a rectangular frame ABCD are AC $= (3x + 5)$ cm and BD $= (2x + 15)$ cm. Find x, the length of each diagonal, OA (O is the centre) and the length of wire needed to cover both diagonals. 5

OR

Find the smallest number which leaves remainder 3 when divided by 6, 8 and 12. 5

27. A school has 2000 students who are to stand in rows so that the number of rows equals the number of students in each row. How many more students are needed to complete the arrangement? What is the number of rows then? 5

OR

Simplify $\frac{3^2 \times 2^{-1}}{3^{-1} \times 2^2}$ and write the answer with positive exponents. Then find its value. 5

END OF QUESTION PAPER **Total: 50 Marks**

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